

ON THE MOMENTS OF TRACES OF WISHART AND INVERTED WISHART MATRICES

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Summary: The moments of traces of Wishart and inverted Wishart matrices are derived by the use of zonal polynomials of a matrix argument and these are tabulated for 7 and 8 degrees for Wishart distribution and up to 6 degrees for inverted one.

1. INTRODUCTION

Let S be a $p \times p$ symmetric random matrix with a Wishart distribution $W_p(\Sigma, n)$ or an inverted Wishart distribution $IW_p(\Sigma, n)$. We consider the moments of monomials for $\text{tr}S, \text{tr}S^2, \dots$ of k degree, i.e.,

$$E_S[(\text{tr}(RS))^{\nu_1}(\text{tr}(RS)^2)^{\nu_2}(\text{tr}(RS)^3)^{\nu_3} \dots],$$

where R is a $p \times p$ symmetric matrix, and $\nu = (1^{\nu_1} 2^{\nu_2} 3^{\nu_3} \dots)$ is a partition of k consisting of ν_1 ones, ν_2 twos, etc. These moments play an important role in studying the distributions of various multivariate statistics. It is not difficult to calculate the moments of small degrees, but it becomes quite complicated for higher degrees. When $S \sim W_p(\Sigma, n)$, Fujikoshi [1970, 1973] tabulated the moments of all the monomials up to 6th degrees, by the use of zonal polynomials of a matrix argument. Hayakawa and Kikuchi [1979] have tabulated the moments of all the monomials up to 8th degrees when S has a multivariate symmetric normal distribution.

In this paper we tabulate the moments of all the monomials up to 6th degrees when $S \sim IW_p(\Sigma, n)$. The calculation is based on the use of zonal

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polynomials of a matrix argument. Further, we extend Fujikoshi's tables to the case of $k = 7(1)10$ when $S \sim W_p(\Sigma, n)$. The tables for $k = 7$ and 8 are given in the Appendix.

2. THE CASE OF $S \sim IW_p(\Sigma, n)$

Let S be a $p \times p$ symmetric random matrix with an inverted Wishart distribution $IW_p(\Sigma, n)$, i.e., $S^{-1} \sim W_p(\Sigma, n)$. Das Gupta [1968] obtained $E[S]$ and $E[S^2]$ in the study of the distribution of discriminant function coefficients. Haff [1979] gave the expressions of $E[s_{ij}]$, $cov[s_{ij}, s_{kl}]$ and $E[SRS]$ as the solution of a nonsingular system of linear equations. Fujikoshi [1987] used $E[(tr(RS))^2]$ and $E[tr(RS)^2]$ in obtaining error bounds for asymptotic expansions of the distribution of the MLE in a GMANOVA. These moments have also been used for the evaluation of risk functions of some estimators (see e.g., Brown [1990]). Here we note that these moments can be calculated by the use of zonal polynomials of S . Let $C_k(S)$ be a zonal polynomial of S corresponding to a partition $\kappa = (k_1, \dots, k_q)$ of k such that $k = k_1 + \dots + k_q$ and $k_1 \geq \dots \geq k_q \geq 0$, where q is an integer not more than k . For the definition of $C_\kappa(S)$, see James [1964] and Muirhead [1982]. It is known (Constantine [1966], Khatri [1966]) that if $n - p - 2k + 1 > 0$,

$$(2.1) \quad E_s[C_\kappa(RS)] = e_\kappa C_\kappa(R\Sigma^{-1}),$$

where R is a $p \times p$ symmetric matrix,

$$e_\kappa^{-1} = (-2)^k \left(-\frac{n}{2} + \frac{p+1}{2} \right) \kappa,$$

$(a)_\kappa = \prod_{i=1}^p (a - \frac{1}{2}(i-1))_{k_i}$ and $(a)_m = a(a+1) \cdots (a+m-1)$. As an alternative of $C_k(S)$, let

$$(2.2) \quad Z_\kappa(S) = f_\kappa C_\kappa(S),$$

where $f_\kappa = (2k)! / (2^k k! \chi_{[2\kappa]}(1))$,

$$\chi_{[2\kappa]}(1) = (2k)! \prod_{i < j}^p (2k_i - 2k_j - i + j) / \prod_{i=1}^p (2k_i + p - i)!.$$

The relationship (2.2) between the $Z_{\kappa}(S)$'s and the monomials of k th degree can be expressed as

$$(2.3) \quad Z_{\kappa}(S) = \sum_{\nu_1+2\nu_2+3\nu_3+\dots=k} h_{\kappa\nu} (tr S)^{\nu_1} (tr S^2)^{\nu_2} \dots,$$

where $\nu = (1^{\nu_1} 2^{\nu_2})$ is a partition of k . Let z_S and b_S be the vectors of all the zonal polynomials $Z_{\kappa}(S)$ and the monomials of k th degree such that

$$\begin{aligned} z_S &= (Z_{(k)}(S), Z_{(k-1,1)}(S), \dots, Z_{(1^k)}(S))', \\ b_S &= ((tr S)^k, (tr S)^{k-2} tr S^2, \dots, tr S^k)'. \end{aligned}$$

For a precise order in z_S and b_S , see James [1964] and Parkhurst and James [1974]. Similarly, let

$$c_S = (C_{(k)}(S), C_{(k-1,1)}(S), \dots, C_{(1^k)}(S))',$$

$$D_e = \text{diag}(e_{(k)}, e_{(k-1,1)}, \dots, e_{(1^k)})$$

and

$$D_f = \text{diag}(f_{(k)}, f_{(k-1,1)}, \dots, f_{(1^k)}),$$

we can write the relationship (2.3) as

$$(2.4) \quad z_S = H_k b_S,$$

where $H_k = [h_{\kappa\nu}]$. The matrix H_k has been tabulated by James [1964] for $k = 1(1)6$ and by Parkhurst and James [1974] for $k = 7(1)12$.

Theorem 2.1 Let S be distributed as $IW_p(\Sigma, n)$. If $n - p - 2k + 1 > 0$, then

$$E_S[b_{RS}] = A_k b_{R\Sigma^{-1}},$$

where

$$A_k = H_k^{-1} D_e H_k = B_k' D_e H_k, \quad B_k = [b_{\kappa\nu}], \quad b_{\kappa\nu} = h_{\kappa\nu} / (f_{(k)} h_{(k)\nu}),$$

Proof. The formulae (2.1) and (2.2) can be written as

$$E_s[c_{RS}] = D_e c_{R\Sigma^{-1}} \text{ and } z_S = D_f c_S,$$

respectively. Therefore,

$$\begin{aligned} E_S[b_{RS}] &= E_S[H_k^{-1} z_{RS}] \\ &= E_S[H_k^{-1} D_f c_{RS}] \\ &= H_k^{-1} D_f D_e D_f^{-1} H_k b_{R\Sigma^{-1}} \\ &= H_k^{-1} D_e H_k b_{R\Sigma^{-1}}. \end{aligned}$$

From the orthogonal properties of H_k due to James [1964], it follows that $H_k^{-1} = B_k'$. This completes the proof.

From Theorem 2.1 it is easily seen that

$$E[(tr RS)] = d_1^{-1} tr(R\Sigma^{-1}),$$

$$E \begin{bmatrix} (tr RS)^2 \\ tr(RS)^2 \end{bmatrix} = d_2^{-1} \begin{bmatrix} n-p-2 & 2 \\ 1 & n-p-1 \end{bmatrix} \begin{bmatrix} (tr R\Sigma^{-1})^2 \\ tr(R\Sigma^{-1})^2 \end{bmatrix},$$

$$E \begin{bmatrix} (tr RS)^3 \\ tr(RS)^2 tr RS \\ tr(RS)^3 \end{bmatrix} = d_3^{-1}$$

$$\begin{bmatrix} (n-p)^2 - 5(n-p) + 2 & 6(n-p-3) & 16 \\ n-p-3 & (n-p)^2 - 4(n-p) + 7 & 4(n-p-1) \\ 2 & 3(n-p-1) & (n-p-1)^2 \end{bmatrix}$$

$$\begin{bmatrix} (tr R\Sigma^{-1})^3 \\ tr(R\Sigma^{-1})^2 tr R\Sigma^{-1} \\ tr(R\Sigma^{-1})^3 \end{bmatrix},$$

where d_k denotes the common denominator of the elements of A_k . Let $A_k = d_k^{-1} G_k$ and $n-p=m$, then we have

$$G_k = m^\ell I + m^{\ell-1} G_k^{(\ell-1)} + \dots + m G_k^{(1)} + G_k^{(0)}.$$

These d_k 's up to $k = 6$ are tabulated in the Appendix 1 and the coefficients of G_k 's for $k = 2(1)6$ are in 2.

We note that the summations of the row elements of A_k become the same value for each k . Namely, if $a_{\kappa\nu}$ denotes the element of A_k , then

$$\begin{aligned}\sum_{\nu} a_{\kappa\nu} &= \sum_{\nu} (\sum_{\mu} b_{\mu\kappa} e_{\mu} h_{\mu\nu}) \\ &= \sum_{\mu} b_{\mu\kappa} e_{\mu} \sum_{\nu} h_{\mu\nu} \\ &= b_{(k)} e_{(k)} f_{(k)} \\ &= e_{(k)} \\ &= 1/\{(n-p-1) \cdots (n-p-1-2k+2)\}.\end{aligned}$$

From this fact we can check the accuracy of the calculation.

3. THE CASE OF $S \sim W_p(\Sigma, n)$

When $S \sim W_p(\Sigma, n)$, it is known (Fujikoshi [1973]) that

$$(3.1) \quad E_S[b_{RS}] = 2^{-k} F_k b_{R\Sigma},$$

where

$$F_k = H_k^{-1} D_k H_k,$$

$$D_k = 2^k \text{diag} \left(\left(\frac{n}{2} \right)_{(k)}, \left(\frac{n}{2} \right)_{(k-1,1)}, \dots, \left(\frac{n}{2} \right)_{(1^k)} \right).$$

Letting $D_k = n^k I + n^{k-1} D_k^{(k-1)} + \dots + n D_k^{(1)}$ and noting that $H_k^{-1} = B_k'$, we have

$$(3.2) \quad F_k = n^k I + n^{k-1} F_k^{(k-1)} + \dots + n F_k^{(1)},$$

Hayakawa and Kikuchi [1979] noted that

$$\sum_{\mu} (F_k)_{\kappa\mu} = n(n+2) \cdots (n+2k-2).$$

The matrices $F_k^{(i)}$ have been tabulated by Fujikoshi [1973] up to $k = 6$. We extend his table to the case of $k = 7(1)10$. Here we tabulate $F_k^{(i)}$ for $k = 7, 8$ in the Appendix 3.

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APPENDIX: Tables of d_k , $G_k^{(i)}$ and $F_k^{(i)}$

1. Tables of d_k for $k = 1(1)6$

$$d_1 = (m-1)$$

$$d_2 = m(m-1)(m-3)$$

$$d_3 = (m+1)m(m-1)(m-3)(m-5)$$

$$d_4 = (m+2)(m+1)m(m-1)(m-2)(m-3)(m-5)(m-7)$$

$$d_5 = (m+3)(m+2)(m+1)m(m-1)(m-2)(m-3)(m-5)(m-7)(m-9)$$

$$d_6 = (m+4)(m+3)(m+2)(m+1)m(m-1)^2(m-2)(m-3)(m-4)(m-5)(m-7)(m-9)(m-11)$$

2. Tables of $G_k^{(i)}$ for $k = 2(1)6$

$$G_2^{(0)} = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}, G_3^{(0)} = \begin{pmatrix} 2 & -18 & 16 \\ -3 & 7 & -4 \\ 2 & -3 & 1 \end{pmatrix}, G_3^{(1)} = \begin{pmatrix} -5 & 6 & 0 \\ 1 & -4 & 4 \\ 0 & 3 & -2 \end{pmatrix},$$

$$G_4^{(0)} = \begin{pmatrix} -32 & -48 & 288 & 320 & -528 \\ -4 & 92 & -48 & -128 & 88 \\ 24 & -48 & 36 & -16 & 4 \\ 10 & -48 & -6 & 68 & -24 \\ -11 & 22 & 1 & -16 & 4 \end{pmatrix},$$

$$G_4^{(1)} = \begin{pmatrix} 8 & 216 & -84 & -384 & 240 \\ 18 & -84 & 62 & 128 & -128 \\ -7 & 62 & 29 & -112 & 24 \\ -12 & 48 & -42 & -40 & 42 \\ 5 & -32 & 6 & 28 & -11 \end{pmatrix},$$

$$G_4^{(2)} = \begin{pmatrix} 28 & -108 & 12 & 64 & 0 \\ -9 & 37 & -16 & -56 & 40 \\ 1 & -16 & 15 & 32 & -36 \\ 2 & -21 & 12 & 27 & -24 \\ 0 & 10 & -9 & -16 & 11 \end{pmatrix}, G_4^{(3)} = \begin{pmatrix} -11 & 12 & 0 & 0 & 0 \\ 1 & -10 & 2 & 8 & 0 \\ 0 & 2 & -9 & 0 & 8 \\ 0 & 3 & 0 & -8 & 6 \\ 0 & 0 & 2 & 4 & -5 \end{pmatrix}$$

$$G_5^{(0)} = \begin{pmatrix} -128 & -1840 & -3360 & 4160 & 14080 & 1680 & -14592 \\ -92 & 68 & 1392 & 560 & -1760 & -1992 & 1824 \\ -56 & 464 & -660 & -16 & 544 & -372 & 96 \\ 52 & 140 & -12 & -988 & -104 & 1464 & -552 \\ 88 & -220 & 204 & -52 & -68 & 60 & -12 \\ 7 & -166 & -93 & 488 & 40 & -372 & 96 \\ -38 & 95 & 15 & -115 & -5 & 60 & -12 \end{pmatrix},$$

$$G_5^{(1)} = \begin{pmatrix} -408 & 760 & 2580 & 3200 & -2880 & -8640 & 5376 \\ 38 & 368 & -810 & -936 & 2120 & 1704 & -2496 \\ 43 & -270 & 363 & 712 & -8 & -1140 & 288 \\ 40 & -234 & 534 & 334 & -962 & -516 & 792 \\ -18 & 265 & -3 & -481 & 69 & 180 & -24 \\ -36 & 142 & -285 & -172 & 120 & 435 & -216 \\ 14 & -130 & 45 & 165 & -10 & -135 & 39 \end{pmatrix},$$

$$G_5^{(2)} = \begin{pmatrix} 84 & 860 & -720 & -1760 & 320 & 1200 & 0 \\ 43 & -209 & 354 & 508 & -400 & -984 & 672 \\ -12 & 118 & -30 & -352 & 312 & 428 & -480 \\ -22 & 127 & -264 & -163 & 318 & 300 & -312 \\ 2 & -50 & 117 & 159 & -40 & -300 & 96 \\ 5 & -82 & 107 & 100 & -200 & -98 & 152 \\ 0 & 35 & -75 & -65 & 40 & 95 & -46 \end{pmatrix},$$

$$G_5^{(3)} = \begin{pmatrix} 59 & -280 & 60 & 160 & 0 & 0 & 0 \\ -14 & 75 & -78 & -144 & 40 & 120 & 0 \\ 1 & -26 & 52 & 40 & -88 & -76 & 96 \\ 2 & -36 & 30 & 61 & -22 & -108 & 72 \\ 0 & 5 & -33 & -11 & 50 & 60 & -72 \\ 0 & 10 & -19 & -36 & 40 & 44 & -40 \\ 0 & 0 & 15 & 15 & -30 & -25 & 24 \end{pmatrix},$$

$$G_5^{(4)} = \begin{pmatrix} -16 & 20 & 0 & 0 & 0 & 0 & 0 \\ 1 & -15 & 6 & 12 & 0 & 0 & 0 \\ 0 & 2 & -14 & 0 & 8 & 8 & 0 \\ 0 & 3 & 0 & -13 & 2 & 12 & 0 \\ 0 & 0 & 3 & 1 & -12 & 0 & 12 \\ 0 & 0 & 2 & 4 & 0 & -10 & 8 \\ 0 & 0 & 0 & 0 & 5 & 5 & -6 \end{pmatrix},$$

$$G_6^{(i)} = [G_{61}^{(i)} \ G_{62}^{(i)}]$$

$G_{61}^{(0)}$	7232	-89280	-377280	-25600	-811200	691200
	-2976	-23456	-21408	75776	81120	178688
	-2096	-3568	25584	25088	-57744	20992
	-160	14208	28224	-31936	66336	-153600
	-6760	20280	-86616	88448	-80184	218112
	720	5584	3936	-25600	27264	-45184
	1072	-1984	-7680	-6592	-29232	52672
	764	-2292	-11772	18752	-29100	66048
	1952	-5856	13176	-9760	14856	-38496
	-468	-1940	-4380	14000	-60	3920
	-622	1866	702	-3424	6	-480

$G_{62}^{(0)}$	771840	488960	2810880	-1078272	-2388480
	-47616	-48896	-281088	-148992	238848
	-30720	-41856	105408	-56064	14976
	-29664	75008	-87840	201600	-82176
	-175392	-155200	178272	-1152	192
	39504	44032	-57744	9408	-1920
	21552	-41856	53136	-56064	14976
	-47088	-44320	49968	-1152	192
	26568	22208	-23688	-1152	192
	-17520	-320	-720	9408	-1920
	2808	32	72	-1152	192

$G_{61}^{(1)}$	-34400	-16800	169920	650880	428160	977280
	-560	41456	55200	-91200	-123936	-349632
	944	9200	-80280	-1056	24120	149408
	4068	-17100	-1188	-24736	15084	85728
	3568	-30984	36180	20112	-63996	46128
	1018	-10926	28014	14288	5766	-40816
	-1874	1158	2274	30688	4698	-37568
	-2308	9216	-5670	-7784	-10758	17448
	-1855	11421	-6849	-13904	10755	-8160
	-128	3860	-3570	-16680	-6930	15000
	942	-4692	423	8180	687	-1188

$G_{62}^{(1)}$	-1349280	-1477120	-2671200	-294912	3617280
	27792	196608	548208	296448	-600576
	9096	-20160	-54792	-45696	9024
	138096	-31136	-125136	-240192	196320
	28188	-57376	129060	-133056	21984
	-28176	11632	-12240	36000	-4752
	-71160	9936	14376	90528	-43248
	11178	-576	13014	-28512	4560
	7188	5784	-16260	14256	-2568
	28290	-7920	8910	-28944	7920
	-8109	760	-963	4752	-984

$G_{61}^{(2)}$	3600	163440	39960	-301120	-96360	-998400
	5448	-18400	-50892	4864	52452	265536
	222	-8482	4558	12752	-7710	-120672
	-1882	912	14346	24836	-32532	-94344
	-803	13113	-11565	-43376	27051	-22272
	-1040	8298	-22626	-15724	-2784	46724
	-197	2796	-12347	-14072	9792	38888
	648	-8868	7956	13504	9678	-28440
	619	-7422	6834	16012	-5027	932
	614	-3665	9255	8225	620	-17045
	-362	3912	-3459	-6860	-755	4098

$G_{62}^{(2)}$	-141840	414720	891360	1414656	-1390080
	67104	-189184	-356256	-281472	500736
	-49388	28288	54672	118464	-73792
	-63324	54016	144108	118440	-164640
	58752	51616	-60324	11904	-24160
	29166	-18960	1398	-40908	16392
	40658	-1552	-45806	-67456	49232
	-1746	13372	-8586	8604	-6186
	-22903	-3816	3451	10656	600
	-21080	2390	6660	25672	-11710
	9231	-1031	225	-7026	1963

$G_{61}^{(3)}$	11760	-52440	-90900	-56320	21000	288000
	-1748	-1680	25224	18720	-11736	-119328
	-505	4204	-11689	-16672	4098	35856
	-352	3510	-18756	-8666	9900	60762
	175	-2934	6147	13200	4752	-21312
	300	-3729	6723	10127	-2664	-22851
	445	-2982	10784	4140	-5296	-22448
	-68	2148	-6327	-6252	2454	4614
	-81	2100	-4980	-6056	-1656	11188
	-192	2225	-4950	-3330	2300	10685
	42	-1212	3165	2612	-168	-4080

$G_{62}^{(3)}$

320400	-43520	-116640	-442368	161280
-71568	45824	100800	170880	-155136
43136	-22496	-39840	-63360	67520
18630	-25008	-54504	-47952	62688
-31776	13088	-19872	44160	-5376
-16836	3076	16782	25644	-16320
-12705	11120	21914	22288	-27008
12510	-5845	504	-5904	2418
10957	224	-2900	-12384	3840
6965	-1640	-7740	-12641	8570
-5064	403	1440	5142	-2028

 $G_{61}^{(4)}$

-3382	-10050	31860	46080	-2520	-38400
-335	3015	-10356	-10768	2352	27936
177	-1726	2245	6400	-1508	-11328
288	-2019	7200	2899	-1260	-15444
-21	588	-2262	-1680	-2909	11424
-40	873	-2124	-2574	1428	5445
-91	1194	-3020	-1848	908	8192
4	-216	1575	1052	-1170	-2928
5	-258	1626	976	98	-4600
14	-515	1470	1200	-800	-3590
0	126	-915	-564	310	2100

 $G_{62}^{(4)}$

-65520	2560	7200	32256	0
28656	-4608	-12384	-39552	16128
-12080	5600	13008	18816	-19520
-8316	4208	8784	17280	-13536
5448	-6240	1176	-15360	9920
6144	-1952	-6900	-8616	8400
3177	-3568	-5948	-6016	7104
-4014	-1239	3744	5112	-1836
-2974	1664	1451	4992	-2896
-1880	1420	3120	3185	-3540
1332	-306	-1086	-2124	1211

$G_6^{(5)}$

-373	5850	-4140	-7040	120	1920	3600	0	0	0	0
195	-1158	2508	2960	-264	-3360	-4608	256	720	2688	0
-23	418	-838	-960	388	2880	1712	-608	-1752	-3072	1792
-44	555	-1080	-990	72	1824	2520	-304	-648	-3312	1344
1	-66	582	96	-60	-1776	-444	512	2532	1344	-2784
2	-105	540	304	-222	-1110	-1068	648	1212	1704	-1968
5	-192	428	560	-74	-1424	-882	320	764	1600	-1168
0	12	-171	-76	96	972	360	106	-1134	-1296	1068
0	15	-219	-72	211	808	382	-504	-596	-1104	1016
0	35	-240	-230	70	710	500	-360	-690	-638	780
0	0	84	56	-87	-492	-219	178	381	468	-432

 $G_6^{(6)}$

229	-750	180	320	0	0	0	0	0	0	0
-25	248	-288	-368	12	160	240	0	0	0	0
1	-48	220	48	-46	-352	-164	32	96	192	0
2	-69	54	216	0	-132	-360	16	36	216	0
0	3	-69	0	169	96	24	0	-468	0	224
0	5	-66	-22	12	214	66	-80	-114	-204	168
0	10	-41	-80	4	88	180	0	-38	-256	112
0	0	9	4	0	-120	0	122	90	108	-234
0	0	12	4	-39	-76	-19	40	161	96	-200
0	0	15	15	0	-85	-80	30	60	134	-110
0	0	0	0	7	42	21	-39	-75	-66	89

 $G_6^{(7)}$

-27	30	0	0	0	0	0	0	0	0	0
1	-26	12	16	0	0	0	0	0	0	0
0	2	-25	0	2	16	8	0	0	0	0
0	3	0	-24	0	6	18	0	0	0	0
0	0	3	0	-24	0	0	0	24	0	0
0	0	3	1	0	-23	0	4	6	12	0
0	0	2	4	0	0	-21	0	2	16	0
0	0	0	0	0	6	0	-21	0	0	18
0	0	0	0	2	4	1	0	-20	0	16
0	0	0	0	0	5	5	0	0	-17	10
0	0	0	0	0	0	0	3	6	6	-12

3. Tables of $F_k^{(i)} = [F_{k1}^{(i)} \ F_{k2}^{(i)}]$ for $k = 7, 8$ $F_{71}^{(1)}$

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1536
0	0	0	0	0	0	0	1152
0	0	0	0	384	1152	384	1152
0	0	0	0	288	864	288	1248
0	0	0	0	192	576	192	1152
0	0	192	96	288	1248	576	1056
0	0	144	72	288	1008	432	1224
0	0	128	64	288	960	384	1248
0	0	80	40	240	720	240	1320
0	24	168	120	288	1296	744	1080
0	20	160	100	300	1320	680	1060
0	12	120	60	300	1080	480	1212
1	21	175	140	287	1260	700	1092

 $F_{72}^{(1)}$

0	0	0	0	0	0	0	46080
0	0	0	7680	7680	7680	23040	
1536	3072	1536	5376	6144	7680	19200	
1152	2304	1152	5760	5760	5760	23040	
1536	3456	2304	4608	5760	9600	15744	
1344	2880	1728	5184	6336	8640	17280	
1152	2304	1152	5952	6528	7680	19200	
1632	3744	3168	4320	5088	9696	14976	
1440	3456	2592	4608	5760	8640	16416	
1536	3552	2496	4640	5696	8960	16128	
1440	3120	1920	5120	6080	8000	17760	
1536	3480	3072	4416	5280	9504	15072	
1600	3560	3040	4400	5200	9680	14960	
1440	3360	2400	4752	5808	8880	16176	
1596	3528	3108	4396	5236	9436	15104	

 $F_{71}^{(2)}$

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	2560
0	0	0	0	576	1984	768	800
0	0	0	0	480	1632	624	1056
0	0	288	160	176	960	576	1392
0	0	248	136	240	1032	528	1520
0	0	192	104	288	1056	432	1624
0	32	100	88	348	1520	812	1288
0	30	120	96	366	1488	804	1416
0	28	128	92	372	1528	792	1308
0	22	140	80	390	1520	720	1360
1	9	191	140	351	1344	668	1436
1	11	195	140	325	1340	660	1380
1	15	195	140	341	1320	660	1464
0	21	175	105	364	1470	735	1393

$F_{72}^{(2)}$	0	0	0	13440	16128	26880	0
	3200	6720	4224	2880	4224	9600	23040
	1280	3072	2688	6112	7488	12480	19200
	1536	3312	2304	6720	8064	13440	17280
	1952	4464	3744	5616	6240	10912	19968
	1984	4584	3648	5376	6432	10560	20160
	2144	4752	3456	5344	6336	10560	20160
	1888	3924	3264	5744	6624	11712	19104
	1752	3960	3168	5736	7056	11448	19008
	1760	3928	3168	5776	6848	11424	19296
	1760	3960	3040	5880	7016	11360	19200
	1912	4332	3528	5480	6696	11160	19200
	1960	4280	3508	5580	6548	11240	19280
	1908	4284	3408	5580	6744	10948	19440
	1848	4221	3360	5600	6748	11340	19068

$F_{71}^{(3)}$	0	0	0	0	0	0	3360
	0	0	0	0	720	2720	400
	0	0	336	200	120	640	432
	0	0	300	176	180	816	432
	0	36	60	60	200	1032	636
	0	34	80	68	258	1180	684
	0	30	108	72	318	1368	720
	1	5	107	92	123	392	212
	1	6	129	104	126	564	324
	1	7	139	108	113	592	356
	1	10	165	120	150	760	460
	0	9	41	15	174	630	261
	0	11	45	30	170	620	305
	0	15	75	60	172	780	435
	0	0	70	35	154	525	210

$F_{72}^{(3)}$	4480	10080	8064	0	0	0	0
	640	1680	1920	2880	4224	9600	0
	1376	3336	3168	1312	1728	4800	7680
	1664	3888	3456	720	1728	5760	5760
	672	1356	1440	2784	3264	5504	8448
	752	1776	1824	2520	2976	6240	7200
	864	2136	2208	2088	2928	6960	5760
	880	1968	1488	2636	3216	4512	9600
	1012	2178	1872	2376	2736	5184	8712
	968	2104	1944	2404	2792	4968	8832
	1040	2430	2308	2040	2480	5600	7800
	792	1803	1224	2760	3336	5112	9168
	760	1745	1240	2780	3368	4980	9260
	864	1863	1680	2556	2988	5520	8412
	847	1932	1365	2674	3241	4907	9324

$F_{71}^{(4)}$

4480	10080	8064	0	0	0	0
640	1680	1920	2880	4224	9600	0
1376	3336	3168	1312	1728	4800	7680
1664	3888	3456	720	1728	5760	5760
672	1356	1440	2784	3264	5504	8448
752	1776	1824	2520	2976	6240	7200
864	2136	2208	2088	2928	6960	5760
880	1968	1488	2636	3216	4512	9600
1012	2178	1872	2376	2736	5184	8712
968	2104	1944	2404	2792	4968	8832
1040	2430	2308	2040	2480	5600	7800
792	1803	1224	2760	3336	5112	9168
760	1745	1240	2780	3368	4980	9260
864	1863	1680	2556	2988	5520	8412
847	1932	1365	2674	3241	4907	9324

 $F_{72}^{(4)}$

0	0	0	0	0	0	0
640	1680	1920	0	0	0	0
192	408	480	640	768	1920	0
96	432	864	240	576	1920	0
320	696	576	336	672	704	1920
360	780	792	288	312	1200	1440
320	846	1080	48	288	1440	960
64	348	120	688	1032	960	2112
210	396	360	638	576	1446	1728
176	385	384	588	728	1376	1632
200	480	640	400	530	1700	1200
212	441	240	659	720	984	2328
160	455	265	600	845	880	2300
241	516	510	492	528	1301	1860
168	357	210	679	805	1085	2247

 $F_{71}^{(5)}$

0	0	420	280	0	0	0	0
0	0	20	20	60	320	240	0
1	2	41	40	12	48	20	56
1	3	60	52	0	36	72	12
0	3	6	0	27	72	24	36
0	4	6	3	18	89	48	8
0	6	5	12	12	96	93	0
0	0	7	1	9	8	0	63
0	0	9	2	0	24	0	12
0	0	8	4	7	16	6	32
0	0	10	10	0	25	30	10
0	0	0	0	6	18	1	11
0	0	0	0	10	15	5	30
0	0	0	0	5	30	15	0
0	0	0	0	0	0	0	21

$F_{72}^{(5)}$

0	0	0	0	0	0	0
0	0	0	0	0	0	0
96	192	192	0	0	0	0
32	144	288	0	0	0	0
0	84	0	96	192	160	0
32	48	60	72	72	240	0
0	36	144	8	48	240	0
6	96	24	44	144	0	288
59	90	72	72	0	144	216
40	103	72	28	56	136	192
40	90	145	0	20	200	120
24	21	0	115	96	120	288
0	35	10	80	155	100	260
24	51	60	60	60	215	180
21	42	21	84	91	105	315

 $F_{71}^{(6)}$

0	42	0	0	0	0	0	0
1	1	20	20	0	0	0	0
0	2	2	0	6	24	8	0
0	3	0	3	0	12	24	0
0	0	3	0	3	0	0	12
0	0	3	1	0	4	0	2
0	0	2	4	0	0	6	0
0	0	0	0	3	2	0	5
0	0	0	0	0	6	0	0
0	0	0	0	2	4	1	0
0	0	0	0	0	5	5	0
0	0	0	0	0	0	0	2
0	0	0	0	0	0	0	5
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

 $F_{72}^{(6)}$

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	24	0	0	0	0	0
8	12	12	0	0	0	0
0	6	24	0	0	0	0
0	0	0	8	24	0	0
6	0	0	12	0	18	0
0	7	0	4	8	16	0
0	0	10	0	2	20	0
4	3	0	9	0	0	24
0	5	1	0	11	0	20
3	6	6	0	0	15	12
0	0	0	7	7	7	21

$F_{81}^{(1)}$

0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	768	9216	4608	9216	3072	
0	0	0	0	0	0	0	576	6912	3456	6912	2304	
0	0	0	0	0	0	0	384	4608	2304	4608	1536	
0	0	0	0	1536	3072	768	384	9216	6144	13824	6144	
0	0	0	0	1152	2304	576	576	8448	4992	10944	4608	
0	0	0	0	864	1728	432	576	7488	4032	8640	3456	
0	0	0	0	768	1536	384	576	7296	3840	8064	3072	
0	0	0	0	480	960	240	480	5760	2880	5760	1920	
0	0	360	144	1152	3168	1080	504	8784	6336	14688	8064	
0	0	320	128	1152	2944	960	480	9216	6336	14592	7552	
0	0	240	96	1008	2304	720	576	8640	5184	12240	5952	
0	0	200	80	960	2080	600	600	8880	5280	12000	5440	
0	0	120	48	720	1440	360	600	7920	4320	9360	3840	
0	32	336	192	1152	3456	1488	496	8640	6144	13920	8192	
0	30	330	180	1170	3480	1440	510	8580	6240	14040	8160	
0	24	300	144	1200	3360	1260	492	8784	6240	14040	7680	
0	14	210	84	1050	2520	840	574	8484	5040	11760	5600	
1	28	350	224	1148	3360	1400	509	8736	6384	14112	8288	

 $F_{82}^{(1)}$

0	0	0	0	0	0	0	0	0	0	0	645120
0	0	0	0	0	0	46080	92160	92160	92160	92160	322560
15360	15360	30720	30720	15360	32256	67584	76800	92160	92160	268800	
11520	11520	23040	23040	11520	34560	69120	69120	69120	69120	322560	
12288	13824	32256	36864	23040	27648	59904	76800	115200	220416		
12672	13248	27648	29952	17280	31104	66816	80640	103680	241920		
11520	11520	23040	23040	11520	35712	73728	80640	92160	268800		
10752	11520	36864	46080	38400	23808	52224	62976	125952	195456		
11712	13824	34560	42624	31680	25920	54912	70272	116352	209664		
12672	14256	31104	38016	25920	27648	59904	74880	103680	229824		
13056	14592	32640	38400	24960	27840	59904	74880	107520	225792		
13440	14160	29760	32640	19200	30720	65280	76800	96000	248640		
10656	12168	35136	42048	37728	25056	52416	65664	122688	197280		
10816	11776	35840	43136	37312	24480	52736	64832	124416	196096		
11712	13440	32352	40128	30720	26496	56448	71808	114048	211008		
11680	13480	33440	40640	30400	26400	56000	71360	116160	209440		
12576	13992	30720	36480	24000	28512	61248	75840	106560	226464		
11136	12240	35328	42240	38016	24576	52992	65664	120576	198304		
10920	12360	35280	42000	38280	24840	52560	65760	120240	198720		
10944	12156	35904	42816	37920	24624	52608	65088	122112	197424		
11928	13608	33264	40656	31080	26376	56112	71232	113232	211456		
10960	12256	35168	41888	37744	24788	52992	65808	120832	198144		

$F_{81}^{(2)}$

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1152	16384	9472	19584	7680
0	0	0	0	0	0	0	960	13440	7680	15840	6144
0	0	0	0	2112	4736	1344	352	7104	5376	12672	6912
0	0	0	0	1776	3936	1104	480	8256	5472	12240	5952
0	0	0	0	1344	2944	816	576	8832	5376	11232	4608
0	0	576	256	704	2560	1152	560	11136	7808	17856	9984
0	0	512	224	888	2752	1104	696	11312	7712	17736	9664
0	0	444	192	1008	2736	1008	732	11568	7392	17208	9024
0	0	408	176	1056	2720	936	744	11824	7648	17472	8832
0	0	300	128	1080	2480	720	780	12080	7520	16560	7680
0	42	210	144	1410	4032	1620	690	10560	7416	15732	8640
0	40	228	144	1440	4064	1604	580	10384	7296	15704	8576
0	38	258	156	1434	3944	1584	702	10668	6912	15552	8416
0	34	270	144	1470	4000	1520	650	10320	6960	15560	8320
0	26	270	120	1470	3760	1320	682	10584	6720	15240	7680
1	12	382	224	1404	3584	1336	689	11488	7648	17328	9408
1	13	385	224	1365	3580	1330	720	11320	7720	17250	9388
1	16	390	224	1316	3560	1320	665	11208	7788	17124	9288
1	21	385	224	1337	3500	1330	728	11340	7476	16758	8988
0	28	350	168	1456	3920	1470	690	11144	7392	16884	8960

 $F_{82}^{(2)}$

0	0	0	0	0	80640	172032	215040	368640	0
26880	28800	72960	78336	49920	17280	39936	61440	138240	322560
8960	10944	33024	41472	34560	36672	78848	99840	168960	268800
11520	12960	37440	41472	28800	40320	86016	107520	184320	241920
15040	17280	45888	56448	47040	33696	69888	84224	146688	279552
16128	18288	47328	57600	45120	32256	68736	85440	144000	282240
17280	19248	50688	58368	42240	32064	68096	84480	145920	282240
14720	13920	44928	49920	43648	33072	73728	85120	159744	264960
14080	15576	42672	49536	42336	34464	72448	89664	155520	267456
14832	16416	41472	50112	40320	34416	74112	93456	153792	266112
13984	15704	41472	49792	40640	34656	73600	91328	153216	270144
14400	16080	42080	49920	38400	35280	75104	92880	154080	268800
14664	16704	45936	53856	46584	32904	69696	87048	152640	265824
14848	15812	46080	53888	46272	32720	71168	85440	153600	266464
15312	17280	44688	54048	44784	32880	70624	89280	148992	268800
14960	16840	45040	53472	44760	33480	70832	87960	149840	269920
15528	17352	44784	53712	42960	33480	71616	89336	147552	272160
14624	16360	43840	53568	44640	32888	70912	88608	153600	263808
14520	16680	44140	53124	44760	33240	70432	88920	153840	263400
14508	16020	44640	52776	44668	33228	71136	87484	154560	264432
14840	16800	43456	52444	43036	33600	71792	90160	151184	266952
14768	16654	44800	53984	45360	32976	70240	88144	152544	264420

$F_{81}^{(3)}$

0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1440	22080	14080	30240	13824
0	0	0	0	2592	6272	1968	240	4800	3840	9504	6144	6144
0	0	0	0	2280	5440	1680	360	6480	4800	10800	5760	5760
0	0	648	304	536	1920	936	400	7920	6080	14472	8832	8832
0	0	588	272	720	2216	936	516	9336	6704	15768	9120	9120
0	0	492	224	936	2496	864	636	10912	7616	17280	9216	9216
0	48	120	96	800	2752	1272	376	3936	2688	5424	3840	3840
0	46	150	108	990	3112	1364	246	4332	3456	7548	4960	4960
0	44	180	120	1140	3352	1428	252	5184	4008	9288	5760	5760
0	42	198	120	1206	3504	1440	226	5032	4128	9616	6016	6016
0	36	240	120	1380	3800	1440	300	6200	4800	11280	6720	6720
1	7	225	152	495	1160	480	384	5832	3652	8082	4224	4224
1	8	246	160	472	1312	568	397	5632	3696	8144	4576	4576
1	9	281	176	597	1792	758	348	6044	4432	9678	5616	5616
1	11	305	184	595	1940	840	340	5900	4440	9830	5948	5948
1	15	345	200	791	2400	1020	344	6648	5028	11574	6888	6888
0	12	82	24	696	1680	522	354	5272	3168	7212	3264	3264
0	13	85	33	690	1670	555	360	5305	3120	7125	3280	3280
0	16	105	60	682	1760	675	357	5148	3144	7098	3600	3600
0	21	175	105	784	2310	945	308	5341	3948	8631	5040	5040
0	0	140	56	616	1400	420	374	5600	3388	7728	3640	3640

 $F_{82}^{(3)}$

35840	40320	107520	129024	107520	0	0	0	0	0
4480	5760	17280	25344	28800	17280	39936	61440	138240	0
9600	11808	35968	46848	44160	7872	17408	26880	76800	107520
12160	14400	42240	52992	48000	4320	12672	28800	86400	80640
5280	5664	19104	22848	25312	16704	35328	43648	86016	118272
4640	5904	21504	27936	28800	15120	32064	42240	95040	100800
5120	6816	24256	32256	32640	12528	28416	43200	103680	80640
8192	9528	22272	26112	22016	16992	36864	45696	67584	143616
7776	8448	24176	28320	26592	15816	33952	41184	73248	134400
7304	8316	26592	31968	30168	14256	29952	37728	81216	121968
7184	8144	25808	31456	30864	14424	30400	37856	78912	123648
7040	8580	28080	35688	34440	12240	26240	36000	86400	109200
7516	8478	20568	24768	18720	17640	37776	47304	75024	137736
7536	8544	20160	24032	18960	17780	37888	47504	73728	138880
6856	7560	22196	26016	23400	16560	35424	43584	80544	128352
6880	7510	21680	25468	23640	16680	35712	43640	79040	129640
6240	7110	23400	28368	27748	15336	32400	40920	85680	117768
7840	8890	22080	26688	20448	17160	36480	46048	73344	138960
7870	8880	22140	26784	20250	17100	36544	45960	73620	138840
7884	8907	21792	26184	20460	17268	36768	46192	72384	139740
7294	8064	23492	27664	24990	16044	34356	42224	77952	130536
7652	8680	21392	25928	19628	17370	37024	46580	74592	138016

$F_{81}^{(4)}$

0	0	0	0	0	0	0	1680	26880	17920	40320	21504
0	0	0	0	3000	7680	2640	120	2400	1920	5040	3840
0	0	716	352	360	1280	672	252	5024	4032	10032	6784
0	0	660	320	540	1680	720	360	6720	5120	12240	7680
0	50	90	72	578	2016	996	98	1656	1248	2580	1920
0	48	120	84	756	2420	1116	72	1584	1512	3708	2880
0	44	168	96	996	2976	1248	60	1840	1920	5136	4096
1	4	126	96	196	384	120	249	2208	1280	2784	1536
1	5	157	112	207	528	230	138	2340	1640	3588	2128
1	6	189	128	234	756	360	120	2460	2092	4518	2832
1	7	209	136	215	824	428	116	2392	2008	4510	3072
1	10	265	160	300	1200	640	120	2660	2360	5760	4108
0	7	30	6	267	560	180	78	1266	402	1440	480
0	8	31	12	254	600	221	155	1244	480	1466	672
0	9	41	15	294	830	291	42	1199	1080	2271	1272
0	11	45	30	290	860	365	70	1350	1020	2315	1480
0	15	75	60	352	1180	555	44	1428	1464	3303	2400
0	0	40	8	84	192	12	117	1376	848	1764	640
0	0	40	11	105	185	30	90	1505	770	1785	665
0	0	45	20	127	230	90	134	1530	721	1881	870
0	0	70	35	154	525	210	70	1540	1267	2772	1645
0	0	0	0	140	280	70	93	1232	672	1428	560

 $F_{82}^{(4)}$

0	0	0	0	0	0	0	0	0	0	0
4480	5760	17280	25344	28800	0	0	0	0	0	0
1280	1488	5248	6912	9600	3840	8192	11520	30720	0	0
480	1080	3600	8640	14400	1440	4224	9600	28800	0	0
2848	3048	9408	11136	11264	2016	5376	8128	16896	26880	0
2240	2700	9960	12384	14160	1728	3552	5520	21600	20160	0
1920	2652	9312	14016	17760	288	1536	5760	23040	13440	0
1664	3720	2688	5376	2816	5616	9216	17152	15360	35712	0
1688	1866	4796	6480	5712	4128	9632	12240	21120	29568	0
804	1080	6480	7344	8784	3816	7392	8652	26064	24192	0
1280	1466	6056	7408	8808	3528	7616	10032	25344	22848	0
800	1260	6360	8880	12000	2400	5320	8700	28200	16800	0
2494	2610	5400	7344	4326	4392	9984	12534	16128	38376	0
2320	3309	5104	7040	4672	4700	8960	13904	14976	38176	0
1800	1914	7068	7824	7512	3954	8152	9168	20976	32592	0
2040	2200	6520	8068	7630	3600	8180	10210	19820	32200	0
1442	1752	7512	9156	10620	2952	6048	7882	24024	26040	0
1952	2350	5272	5760	3936	5025	10112	12800	18624	37392	0
2080	2100	5095	6135	3780	4740	10548	12570	18540	37530	0
2089	2607	4584	6126	3941	4800	10056	13697	17520	37236	0
1568	1638	6118	6937	7007	4074	8652	10220	22344	31458	0
2132	2386	5432	6440	4340	4674	10008	12412	17976	38029	0

$F_{81}^{(5)}$

0	0	0	0	3360	8960	3360	0	0	0	0	0
0	0	780	400	180	640	360	120	2400	1920	5040	3840
0	52	60	48	372	1312	692	24	448	384	816	640
0	50	90	60	540	1760	840	0	180	240	1080	1440
1	3	93	72	85	192	60	54	888	608	1272	768
1	4	123	88	72	262	150	36	762	716	1602	1176
1	6	173	112	60	400	308	24	688	704	1980	1824
0	4	12	0	108	192	48	76	288	0	336	0
0	5	14	3	111	254	86	18	335	96	396	120
0	6	18	6	114	358	132	0	144	306	540	360
0	7	17	12	106	356	165	14	264	240	517	432
0	10	25	30	120	490	285	0	150	300	810	880
0	0	15	2	27	32	0	24	402	107	378	96
0	0	15	4	32	32	6	57	380	112	398	128
0	0	20	5	21	72	6	12	316	284	555	248
0	0	20	10	35	65	30	20	395	220	575	305
0	0	30	20	22	150	90	10	300	361	786	630
0	0	0	0	24	48	2	20	88	96	84	0
0	0	0	0	30	45	5	0	145	60	105	10
0	0	0	0	35	50	15	27	180	24	156	60
0	0	0	0	35	105	35	0	154	168	357	210
0	0	0	0	0	0	0	14	168	84	168	56

 $F_{82}^{(5)}$

0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
640	720	2560	3072	3840	0	0	0	0	0	0
160	360	1200	2880	4800	0	0	0	0	0	0
288	384	672	1344	704	576	1536	2240	3840	0	0
128	144	864	936	1800	432	864	1200	4320	0	0
0	72	192	1152	2880	48	256	960	3840	0	0
512	1368	768	1536	640	960	0	3456	0	5376	0
536	570	1264	1896	1200	264	928	1440	2112	4032	0
214	252	1704	1584	2166	432	576	288	3456	3024	0
336	398	1480	1840	2096	168	448	832	3264	2688	0
160	240	1240	1830	3150	0	80	600	3600	1680	0
413	306	408	864	144	648	1944	2310	2160	5400	0
272	720	288	800	272	884	1280	3024	1920	5056	0
136	108	1031	888	984	690	1304	1200	3768	4032	0
240	250	740	1165	980	480	1260	1750	3500	3640	0
48	102	984	1176	1901	360	720	1030	4440	2520	0
288	442	920	768	480	852	1408	1760	2304	6096	0
405	300	815	945	450	660	1715	1680	2220	6090	0
385	567	600	1050	515	660	1344	2272	1920	5820	0
210	210	1099	1225	1295	504	1036	1120	3507	4410	0
300	316	672	728	420	762	1624	1940	2520	5908	0

$F_{81}^{(6)}$

0	0	840	448	0	0	0	0	0	0	0
0	54	30	24	180	640	360	0	0	0	0
1	2	61	48	24	64	20	12	224	192	384
1	3	90	64	0	60	90	0	60	80	360
0	3	6	0	45	96	24	6	72	0	84
0	4	6	3	36	121	54	0	24	48	72
0	6	5	12	24	144	117	0	0	0	72
0	0	6	0	12	0	0	30	96	0	96
0	0	7	1	9	8	0	6	107	24	108
0	0	9	2	0	24	0	0	36	83	126
0	0	8	4	7	16	6	4	72	56	133
0	0	10	10	0	25	30	0	30	60	150
0	0	0	0	9	8	0	0	30	6	0
0	0	0	0	10	8	1	9	20	0	14
0	0	0	0	6	18	1	0	11	24	21
0	0	0	0	10	15	5	0	30	0	35
0	0	0	0	5	30	15	0	0	24	51
0	0	0	0	0	0	0	4	16	16	12
0	0	0	0	0	0	0	0	25	10	15
0	0	0	0	0	0	0	5	30	3	21
0	0	0	0	0	0	0	0	21	21	42
0	0	0	0	0	0	0	0	0	0	0

 $F_{82}^{(6)}$

0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
256	0	0	0	0	0	0	0	0	0	0
480	0	0	0	0	0	0	0	0	0	0
0	96	120	192	384	160	0	0	0	0	0
60	32	36	216	216	360	0	0	0	0	0
192	0	12	32	192	480	0	0	0	0	0
0	0	216	0	0	0	192	0	640	0	0
24	40	30	44	144	0	48	160	240	288	0
72	12	0	144	0	144	72	96	36	432	0
80	0	14	56	112	136	24	64	112	384	0
185	0	0	0	60	300	0	8	60	360	0
0	89	54	72	144	18	0	192	162	0	504
0	48	150	48	128	32	68	0	304	0	448
0	24	18	175	120	120	54	72	0	288	336
10	40	40	100	199	110	0	44	110	260	280
60	6	12	120	132	275	0	0	30	360	168
0	0	34	72	0	0	126	128	160	192	528
1	40	0	45	33	0	60	219	150	180	510
6	27	57	0	66	15	60	120	290	144	444
21	0	0	84	91	105	42	84	84	399	294
0	28	28	56	56	28	66	136	148	168	574

$F_{81}^{(7)}$

0	56	0	0	0	0	0	0	0	0	0
1	1	30	24	0	0	0	0	0	0	0
0	2	2	0	12	32	8	0	0	0	0
0	3	0	3	0	20	30	0	0	0	0
0	0	3	0	3	0	0	2	24	0	24
0	0	3	1	0	4	0	0	6	12	18
0	0	2	4	0	0	6	0	0	0	12
0	0	0	0	4	0	0	4	0	0	0
0	0	0	0	3	2	0	0	5	0	0
0	0	0	0	0	6	0	0	0	6	0
0	0	0	0	2	4	1	0	0	0	7
0	0	0	0	0	5	5	0	0	0	0
0	0	0	0	0	0	0	0	6	1	0
0	0	0	0	0	0	0	2	4	0	2
0	0	0	0	0	0	0	0	2	4	3
0	0	0	0	0	0	0	0	5	0	5
0	0	0	0	0	0	0	0	0	3	6
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0

 $F_{82}^{(7)}$

0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0
32	0	0	0	0	0	0	0	0	0	0
0	0	48	0	0	0	0	0	0	0	0
0	8	6	8	24	0	0	0	0	0	0
0	2	0	24	0	18	0	0	0	0	0
0	0	2	8	16	16	0	0	0	0	0
10	0	0	0	6	30	0	0	0	0	0
0	7	0	0	0	0	0	24	18	0	0
0	0	8	0	0	0	8	0	32	0	0
0	0	0	9	0	0	6	8	0	24	0
1	0	0	0	11	0	0	4	10	20	0
6	0	0	0	0	15	0	0	2	24	0
0	0	4	8	0	0	12	0	0	0	32
0	5	0	5	3	0	0	13	0	0	30
0	3	6	0	6	1	0	0	16	0	24
0	0	0	7	7	7	0	0	0	21	14
0	0	0	0	0	0	4	8	8	8	28